

S.No. : 613

NBS 4202

No. of Printed Pages : 04

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 49906

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B. Tech. Examination 2023-24

(Even Semester)

ENGINEERING PHYSICS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) What are coherent sources?
- (b) What do you mean by grating element?
- (c) What are the characteristics of a wave function?
- (d) What is Bragg's law?
- (e) Define specific rotation.
- (f) Show that velocity of matter wave is greater than velocity of light.

[P. T. O.]

- (g) What are inertial and non-inertial frames?
(h) Show that rest mass of photon is zero.

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) In Newton's ring experiment, the diameter of 15th dark ring was found to be 0.590 cm and that of 5th ring is 0.336 cm. If the radius of plano-convex lens is 100 cm, calculate the wavelength of light used.
- (b) An electron has speed $4 \times 10^5 \text{ ms}^{-1}$ within the accuracy of 0.01%. Calculate the uncertainty in the position of electron.
- (c) If earth receives $2 \text{ cal min}^{-1} \text{ cm}^{-2}$ solar energy, what are the amplitudes of electric and magnetic field of radiation?
- (d) If the kinetic energy of a body is twice the rest energy, find the velocity of body.

SECTION – C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Discuss the formation of Newton's ring in reflected light. Prove that in reflected light, the diameter of dark ring is proportional to the square root of natural number.
- (b) Describe the Rayleigh's criterion for resolution. Derive an expression for resolving power of grating.
- (c) Explain the construction and working of Nicol prism.
4. (a) Derive time independent Schrodinger wave equation.
- (b) Show that group velocity is equal to the velocity of particle.
- (c) What are matter waves? Show that De-broglie wavelength associated with a particle of mass 'm' and kinetic energy 'E' is given by :

$$\lambda = \frac{h}{\sqrt{2mE}}$$

5. (a) What is Poynting vector? Derive and explain Poynting theorem.

[P. T. O.]

- (b) Show that velocity of plane electromagnetic wave in free space is given by :

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

- (c) Explain the basic principle of optical fibre. Discuss fibre classification.

6. (a) What was the objectives of Michelson-Morley experiment? Describe the experiment. How is the negative result of experiment interpreted?

- (b) Establish mass-energy relation.

- (c) Derive relativistic formula for the variation of mass with velocity.
